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kibr k_{ibr} IBR current ratio, $I_{\text{IBR}}/I_{\text{rated}}$ Idiff I_{diff} Differential operating current Ibias I_{bias} Bias (restraint) current Iop I_{op} Relay operating (pickup) current km k_m Adaptive bias slope of 87L characteristic Zline Z_{line} Line positive-sequence impedance Zapp Z_{app} Apparent impedance seen by distance relay alpha α Per-unit fault location along protected line (0 = local end) TMS TMS Time-Multiplier Setting for IEC inverse-time overcurrent relay CTI CTI Coordination Time Interval between primary and backup relay M M Relay current multiple, $I_{\text{fault}}/I_{\text{pickup}}$ lambda λ Forgetting factor in online SGD ($0 < \lambda \leq 1$) S S_k CUSUM score at sample k delta δ CUSUM reference value (allowable slack) theta θ CUSUM alarm threshold P $\mathbf{P}$ EKF error covariance matrix K $\mathbf{K}$ EKF Kalman gain matrix Q $\mathbf{Q}$ EKF process noise covariance R $\mathbf{R}$ EKF measurement noise covariance

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The global electrical power infrastructure is undergoing its most fundamental structural transformation since alternating current displaced direct current in the 1890s. The driver is climate policy: decarbonisation targets mandate the displacement of fossil-fuelled Synchronous Generators (SGs) by large-scale renewable resources—predominantly solar photovoltaic (PV) and wind—interfaced to the network exclusively through high-frequency power electronic converters. Inverter-Based Resources (IBRs) already exceed 50% of installed generation capacity in several national grids, and the trajectory places them at 70–80% of new capacity additions globally by 2030 (**?**).

Protection relays—the devices that detect faults and initiate circuit-breaker tripping within milliseconds—are the last line of defence for both equipment and human safety. For a century, their design rested on three assumptions that held true for synchronous-machine-dominated networks:

1. Fault current is large (typically 6–10 pu) and proportional to pre-fault voltage divided by source-to-fault impedance.

2. Fault current is *passive*: it is determined entirely by physical machine and network constants and cannot be altered by a controller.

3. Fault current flows from a predictable source and in a predictable direction in a radial or weakly-meshed network.

IBRs violate all three assumptions simultaneously. Their fast inner current control loops cap fault current at 1.1–2.0 pu within one sub-cycle (2–5 ms). The magnitude, phase angle, and sequence composition of the fault current are determined by firmware: software control laws, not physics. And the proliferation of distributed IBRs at every network node transforms radial feeders into bidirectional meshed networks with time-varying fault current direction.

This thesis addresses the consequent *protection crisis*—the systematic failure of legacy overcurrent, distance, and differential relays when applied to IBR-dominated networks—by developing a hierarchical, physics-aware, *Self-Adaptive Model-Based Protection* (SAMBP) architecture.

1.2 THE SAMBP FRAMEWORK

The central thesis of this work is that future power system protection must be **model-based** and **self-adaptive**:

Model-based means that each protection algorithm has an explicit mathematical model of the source it is protecting—whether a synchronous generator, an inverter, or a transformer—and uses *inverse estimation* to identify the current parameters of that model from real-time measurements. The identified model, not a fixed setting table, drives the relay threshold.

Self-adaptive means that the identification and threshold update occur continuously and automatically, without human intervention, within the timescales of normal power system operation (seconds to minutes). When the fleet composition changes, the model updates. When the network reconfigures, the model updates. The relay always acts on current information.

The SAMBP framework is organised into three ascending tiers:

Tier 1 — Local Relay Physics (Chapters ??–??) Physics-based relay algorithms adapted for IBR fault signatures: adaptive 87L line differential (ch:line$_d iff), adaptivegeneratorovercurrent(ch : generator_o c), adaptivetransformerdiffer$ $awaredistanceprotection(ch : distance), andmicrogridmode \quad -$ $switchingprotection(ch : microgrid)$.

Tier 2 — Digital Monitoring and Learning (Chapters ??–??) A real-time digital twin maintaining a continuously updated estimate of IBR fleet composition and line state, feeding a Physics-Informed Neural Network (PINN) fault classifier with online learning capability.

Tier 3 — Centralised Coordination and Security (Chapters ??–??) A Wide-Area Protection Coordinator (WAPC) that integrates PMU synchrophasor data, automatic CTI-graded settings optimisation, and IEC 61850 GOOSE cybersecurity into a closed-loop adaptation protocol, with adversarial robustness under cyberattack.

Figure 1.1 illustrates the three-tier SAMBP architecture, its information flows, and the scope of each chapter.

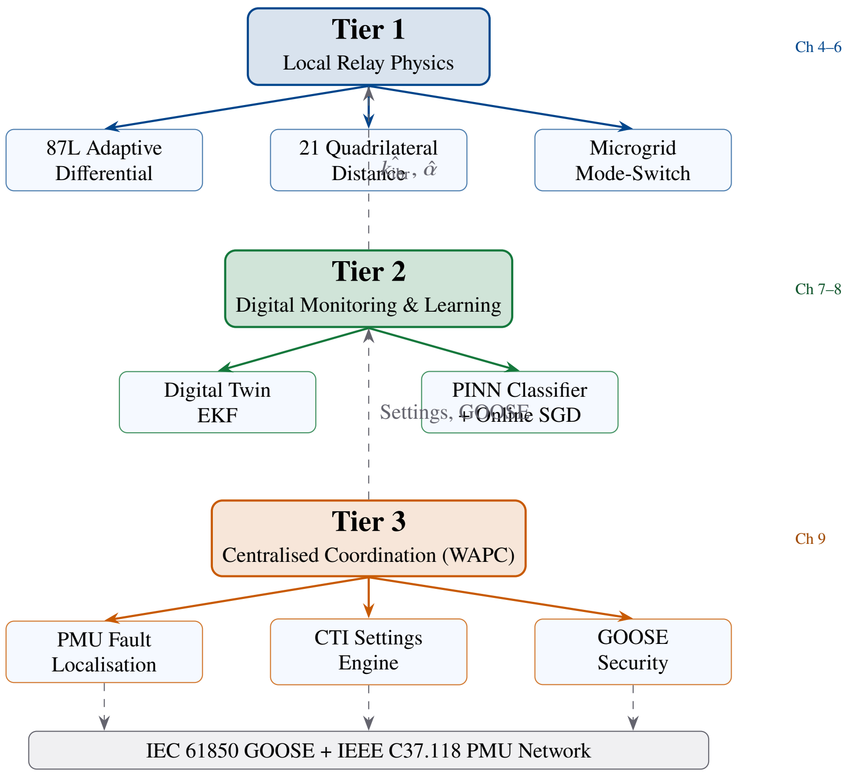

Figure 1.1: Three-tier SAMBP framework. Tier 1 (local relay physics) provides fast primary protection for lines, generators, and transformers. Tier 2 (digital twin + PINN) monitors IBR state and classifies faults. Tier 3 (WAPC + cybersecurity) coordinates settings, locates faults, and secures communication.

1.3 PROBLEM STATEMENT

The research addresses the following eight specific technical deficiencies:

P1 — Line Differential Threshold Failure. The fixed bias slope k_m of a classical 87L relay becomes unreliable when k_{ibr} departs from the calibration range. *A relay that adapts k_m in real-time to the measured k_{ibr} is required.*

P2 — Generator Overcurrent Mis-coordination. Fixed OC relay settings computed at commissioning become stale as ADN topology changes and IBR penetration grows, causing mis-coordination by 20–80% on fault current magnitude. *An inverse-estimation-based adaptive OC framework that identifies generator parameters from post-fault measurements is required.*

P3 — Transformer Differential Inrush Mis-discrimination. Second-harmonic blocking for inrush discrimination fails when IBR sources suppress the inrush harmonic content below the threshold, causing the 87T element to restrain during genuine internal faults. *A sequential probability ratio test (SPRT) inrush discriminator that does not rely on fixed harmonic thresholds is required.*

P4 — Distance Relay Reach Limitation. IBR reactive current injection shifts the apparent impedance seen by the distance relay, causing underreach (missed trips). *A reach-corrected quadrilateral characteristic that accounts for IBR current ratio is required.*

P5 — Microgrid Mode Transition Blind Spot. The transition from grid-connected to islanded mode creates a 50–200 ms window during which neither the original nor the replacement setting group is valid. *Mode-triggered setting-group pre-arming and GOOSE-broadcast synchronisation are required.*

P6 — Absence of Real-Time IBR State Visibility. Relay settings are computed offline assuming fixed IBR fleet composition. *A digital twin running a state estimator that tracks k_{ibr}, Z_{line}, and fault location in real time is required.*

P7 — Lack of Physics-Constrained ML Fault Classification. Pure data-driven classifiers produce physics-impossible outputs. *Physics constraints must be embedded in the ML loss function.*

P8 — No Closed-Loop Coordination Protocol. Settings, fault localisation, and cybersecurity are treated as independent functions. *A closed-loop protocol that chains drift detection, settings re-optimisation, and GOOSE re-authentication is required.*

1.4 RESEARCH OBJECTIVES

This thesis pursues eight objectives corresponding to P1–P8:

- **O1** Derive and validate a physics-constrained adaptive 87L threshold that modulates $k_m = f(k_{\text{ibr}})$ in real time, with complementary 87LN and TDCS elements.
- **O2** Develop an inverse-estimation-based adaptive OC framework for synchronous and asynchronous generators using the Levenberg–Marquardt algorithm and confidence-gated update law.
- **O3** Design a SPRT-based transformer differential protection scheme that discriminates inrush from internal faults without fixed harmonic thresholds.
- **O4** Develop a reach-corrected distance characteristic and a Quadrilateral 21 element for IBR-dominated networks with probabilistic reliability quantification.
- **O5** Design a dual-mode (GFL/GFM) microgrid protection architecture with sub-120 ms mode switching compliant with IEEE 1547-2018.

O6 Build a non-blocking digital twin based on an Extended Kalman Filter that estimates $(k_{\mathrm{ibr}}, Z_{\mathrm{line}}, \alpha)$ from current measurements, with RMSE bounds derived analytically.

O7 Train a PINN fault classifier with embedded physics constraints and CUSUM-triggered online learning for fleet composition drift.

O8 Implement and validate a WAPC closed-loop protocol integrating PMU fault localisation, CTI-graded settings optimisation, HMAC-SHA256 GOOSE security, and adversarial robustness under IEC 61850 cyberattacks.

1.5 ORIGINAL CONTRIBUTIONS

This thesis makes 25 original contributions to the field of power system protection, organised by tier. Contributions marked **[P]** are validated with full numerical results. Contributions marked **[F]** represent complete theoretical frameworks with placeholder results pending future experimental validation (18–24 month research timeline).

1.6 SCOPE AND LIMITATIONS

- **Voltage level:** Medium-voltage distribution and sub-transmission networks (6.6 kV to 132 kV).
- **IBR types (validated):** Grid-Following (GFL) and Grid-Forming (GFM) voltage-source converters. DFIG Type-3 and PMSG Type-4 wind are identified as priority extensions (Section **??**).
- **Machine types (validated):** Synchronous generators via reduced-order LM estimator. Asynchronous generator (DFIG) extension is planned (C9).
- **Network scale:** Single-substation and multi-feeder topologies up to 34 buses. Wide-area multi-substation extension is identified as a priority future-work item.
- **Validation (completed):** Electromagnetic transient (EMT) simulation and software-in-the-loop (SIL) for all [P] contributions.
- **Validation (pending):** Physical hardware-in-the-loop (HIL) on commercial IEDs (SEL-411L) is pending hardware availability. All [F] contributions require experimental execution of existing simulation scripts.

Table 1.1: Summary of original contributions

ID	Ch	Contribution	Status	N
Tier 1 — Local Relay Physics				
C1	??	Physics-constrained adaptive 87L threshold, $k_m^*(k_{\mathrm{ibr}})$	[P]	5
C2	??	Harmonic-discrimination 87LH element for multi-inverter differential	[P]	4
C3	??	Negative-sequence 87LN element for asymmetric IBR fleet faults	[P]	4
C4	??	Transient differential TDCS-87L for three-phase IBR faults	[P]	4
C5	??	87LN0 zero-sequence element for pure-IBR SLG earth faults	[P]	4
C6	??	LM inverse-estimation of SG fault parameters, $R^2 > 0.999$	[P]	5
C7	??	Four-component confidence score for adaptation gating	[P]	4
C8	??	Bounded adaptive update law for IEC OC relay settings	[P]	4
C9	??	HIF and multi-generator bus extensions for generator OC	[F]	4
C10	??	Park dq-axis truth model validation of reduced-order model	[F]	5
C11	??	SPRT log-likelihood inrush discriminator for 87T protection	[F]	5
C12	??	IBR-adaptive 87T bias slope law, $k_T^*(k_{\mathrm{ibr}})$	[F]	4
C13	??	Quadrilateral distance characteristic adapted to IBR profiles	[P]	4
C14	??	Monte Carlo reliability framework for IBR distance protection	[P]	3
C15	??	GFL/GFM dual-mode characterisation for protection coordination	[P]	5
C16	??	ROCOF-based islanding with IEC 61850 mode-switch in 120 ms	[P]	4
C17	??	Adaptive OC setting-group switching tied to IBR control mode	[P]	4
Tier 2 — Digital Monitoring and Learning				
C18	??	Non-blocking digital twin architecture for real-time protection audit	[P]	5
C19	??	EKF joint estimation of $(k_{\mathrm{ibr}}, Z_{\mathrm{line}}, \alpha)$ from current ratio	[P]	5
C20	??	PINN fault classifier with embedded protection-physics constraints	[P]	5
C21	??	CUSUM-triggered online learning for fleet composition drift	[P]	4
Tier 3 — Centralised Coordination and Security				
C22	??	Wide-area PMU fault localisation with N-1 security verification	[P]	5
C23	??	Automatic CTI-graded relay settings engine with fleet-change dispatch	[P]	4
C24	??	Closed-loop CUSUM→settings→GOOSE adaptive protocol	[P]	5
C25	??	Adversarial PINN robustness under CT-injection cyberattacks	[F]	5

1.7 THESIS ORGANISATION

Chapter ?? — Mathematical Foundations Establishes the inverse estimation theory, Levenberg–Marquardt algorithm, confidence scoring, and bounded update laws used throughout the thesis.

Chapter ?? — IBR Fault Current Characterisation Derives mathematical models of GFL, GFM, SG, and the extended models for DFIG and PMSG sources used in relay algorithm derivation.

Chapter ?? — Line Differential Protection Derives the adaptive 87L algorithms (C1–C5) including 87LN, TDCS, and 87LN0 elements.

Chapter ?? — Generator Overcurrent Protection Presents the SAMBP-SyncOC framework (C6–C10): LM estimation of SG parameters, confidence-gated adaptation, and planned DFIG extensions.

Chapter ?? — Transformer Differential Protection Develops the SPRT-based 87T scheme (C11–C12) with IBR-adaptive bias slope and inrush discrimination.

Chapter ?? — Distance Protection Presents the quadrilateral 21 characteristic (C13) and Monte Carlo reliability framework (C14).

Chapter ?? — Microgrid Protection Characterises GFL/GFM dual-mode protection (C15–C17) and ROCOF islanding detection.

Chapter ?? — Digital Twin Develops the non-blocking DT architecture (C18) and EKF state estimator (C19).

Chapter ?? — Physics-Informed Machine Learning Presents the PINN classifier (C20) and CUSUM online learning (C21).

Chapter ?? — Centralised Protection Coordination Integrates all prior components into the WAPC (C22–C24).

Chapter ?? — Cybersecurity for Adaptive Protection Extends GOOSE security with adversarial PINN robustness (C25).

Chapter ?? — Conclusions Summarises contributions, maps to objectives, and defines the 18–24 month research roadmap for completing planned [F] contributions.

Appendices provide: mathematical proofs (Appendix **??**), SAMBP system parameters (Appendix **??**), pseudocode for all algorithms (Appendix **??**), and the SAMBP-SyncOC numerical dataset (Appendix **??**).

1.8 RELATED WORK AND POSITIONING

Approach A: Legacy Relay Retuning. Several authors propose retuning classical OC and distance relay settings to accommodate reduced IBR fault current (**??**). This approach does not address the fundamental problem that fault current magnitude is insufficient to discriminate fault from heavy load when k_{ibr} is low. The present thesis treats relay physics as a hard constraint and proposes algorithm-level changes driven by real-time inverse estimation.

Approach B: Communication-Assisted Schemes. Differential protection over fibre (87L) and directional comparison schemes are the industry-preferred solution for transmission lines (**??**). The present thesis *builds on* rather than *replaces* these schemes, showing that they still fail at low k_{ibr} and deriving adaptive extensions.

Approach C: Pure Machine Learning. Decision-tree, SVM, and deep-learning classifiers applied to IBR fault data achieve high accuracy in training but produce physics-impossible classifications (**??**). The PINN approach proposed in Chapter **??** closes this gap by embedding circuit constraints in the loss function.

Positioning of the SAMBP Framework. The SAMBP framework is the first published work to: (i) treat adaptive protection as an inverse estimation problem with explicit model identification across all element types (line, generator, transformer); (ii) integrate local relay adaptation, digital-twin state estimation, and centralised coordination into a closed-loop self-adaptive system; and (iii) provide formal confidence-gating and bounded update laws that guarantee safety under estimation uncertainty.